

National Exam of Higher National Diploma-New program – 2021 Session

Spécialty/option : Software Engineering

Paper : Case Study

Duration : 6 hours

Credits : 14

Instructions: Answer all questions

SECTION A: ALGORITHM AND PROGRAMMING (50 marks)

ALGORITHMS. (20 marks)

1. What is an algorithm? List four characteristics of algorithms - **(5marks).**
2. The algorithm below is designed to determine the average weight of two rabbits.
 1. Start
 2. Get weight of first and second rabbits, W1, W2
 3. If $W1 < 0$ OR $W2 < 0$
 4. Display weight cannot be negative
 5. Else
 6. $Average = (W1 + W2) / 2$
 7. Displace W1, W2, Average
 8. End
 - a) How many control structures are used in the algorithm? Name them and identify their line numbers **(4marks)**
 - b) Use a flow chart to represent the algorithm given above **(7marks)**
 - c) Differentiate between an algorithm and a computer program? **(4marks)**

Event Programming (VB.net) = (5 marks)

1. Write VB.NET code to declare a variable to store the age of a person. **(3marks)**
2. Write VB.NET code to prompt a user to input his/her name and then the output will be shown as an example below: Hello John! **(2 marks)**

I- Structural programming [15marks]

1. In the C programming language, data types refer to an extensive system used for declaring variables of different types.

- a) Define a C variable? (2 marks)
- b) State how a variable is defined in C programming. (2 marks)
- c) State two ways in which constants are being defined in C programming. (2 marks)
2. What is a data type as used in C programming? List four data types in C programming. (5 marks)
3. What is the main difference between call by value and call by reference? (1 mark)
4. Evaluate the following using operator precedence in C to come out with the output. (8 marks)

```
#include <stdio.h>
main()
{
    int a = 20;
    int b = 10;
    int c = 15;
    int d = 5;
    int e;
    e = (a + b) * c / d;
    printf("Value of (a + b) * c / d is : %d\n", e);
    e = ((a + b) * c) / d;
    printf("Value of ((a + b) * c) / d is : %d\n", e);
    e = (a + b) * (c / d);
    printf("Value of (a + b) * (c / d) is : %d\n", e);
    e = a + (b * c) / d;
    printf("Value of a + (b * c) / d is : %d\n", e);
    return 0;
}
```

Object Oriented Programming (C++ or JAVA) (10 marks)

1. A certain University is made up of Departments. The departments on their part have a name, programs, lecturers, students. **Implement a C++ or JAVA program** which takes University as a class and Departments as object with name, programs, lecturers, students as members. Initialize name as IT, program as HND, lecturer as Mr. P, student as Epie. The program should be such that it outputs the complete information of the University. (10 marks)

SECTION B: Database Development and Administration. (20marks)

1. Think of an organizational database in which some of the fields in the CUSTOMER table must have the given data types. Explain the meaning of the following data types in the Customer table:
 - i) Customer ID (auto numeric field)

- ii) Customer Name (text field)
- iii) Fee Paid (decimal field)
- iv) Pay Date (date field). (2 x 4 = 8 marks)

2. Table 1 below contains sample data for vehicles and for operators who ply these vehicles. In discussing these data with users, we find that vehicle ID (but not descriptions) uniquely identify vehicles and that operator names uniquely identify operators.

VehicleID	Description	Operator	Route	Tarrif Per Mile
V1	Luxury	Polax	Grand Trail	100
		Ubet	East Route	150
V2	Comfort	Polax	Grand Trail	45
		Ubet	East Route	60
		Minim	South Trunk	35

Table 1: Sample Data for Vehicles and Operations

- a) Convert this table to a relation (named VEHICLE OPERATOR) in first normal form. Illustrate the relation with the sample data in the table.(2 marks)
- b) List the functional dependencies in VEHICLEOPERATOR and identify a candidate key.(3 marks)
- c) For the relation VEHICLE OPERATOR, identify each of the following: an insert anomaly, a delete anomaly, and a modification anomaly.(3 marks)
- d) Draw a relational schema for VEHICLE OPERATOR and show the functional dependencies.(3 marks)
- e) In what normal form is this relation?(1 mark)

SECTION C: WEB DESIGN (54mins, 15 marks, 15%)

- 1. List and briefly explain all the new form elements use in HTML5(2 marks)
- 2. CharSet is a new met a tag attribute in HTML5 which configures the character encoding. Give the tag for this meta tag char set (1 mark)
- 3. List and briefly explain all html elements that supports media content (4 marks)
- 4. Define canvas in the context of html and give its default border size (2 marks)
- 5. Differentiate between CSS and CSS3 (1 marks)
- 6. List and briefly explain the different types of CSS mostly used in web development(3 marks)
- 7. What is the difference between <window.onload> and <onDocumentReady>? (2 marks)

SECTION D: NETWORKING 15marks)

- 1) Explain the importance of layering in a network (2marks)
- 2) Differentiate between each of the following
 - i. OSI model and TCP/IP model (2marks)
 - ii. UDP and TCP (3marks)
- 3) With details describe the data encapsulation process in OSI (2marks)
- 4) For each of the following network give a sketch, two advantages ,two disadvantages and a network protocol or technology employing it (6marks)
 - i. Ring topology
 - ii. Star topology